

FTC Note Scout

Scouting & Strategy Web App

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The Concept:

Students participating in the FIRST Tech Challenge robotics competition could use a tool to collect and curate data about their opponents. This dynamic app allows teams to "scout" their opponents during a live competition, which could improve their decision making around game play and alliance selection. The app pulls data from the FIRST Events API, but users will be able to add their own notes about match performance, robot functionality, or other details that might help them plan for matches.

Tech Stack (PERN):

- **Frontend (The Client):** React + Vite and CSS - UI design and Sends the HTTP requests to *the database server (made by me)*.
- **Backend (The Server):** Node.js with Express to handle Authorization (JWT), talk to the FTC API and the database which caches data from FTC API and records data input by users.
- **Database (The Storage):** PostgreSQL to store user account data, scouting notes, and cached API data.
- **API:** the FTC Events API (<https://ftc-events.firstinspires.org/services/API>) for the foundational team/match data (More information here: <https://ftc-events.firstinspires.org/api-docs/index.html>)
- **Security:** Bcrypt + JSON Web Tokens - secure login and session management (authorization and authentications)

Application Type:

- The project is focused on creating a Fullstack Web app. Attention will be paid to both the front and back end aspects of this app.

Software Requirements and Success Criteria:

Technical & Architectural Success:

- **Secure Authentication:** Implement a robust user creation / login system where user passwords are encrypted using bcrypt and sessions are managed via JSON Web Tokens (JWT).
- **API Integration:** Successfully fetch and parse data from the FTC Events API via the Node.js backend (rather than the frontend) to protect API credentials.
- **Database Integrity:** Establish a PostgreSQL schema with Foreign Key relationships (e.g., ensuring a ScoutingNote cannot exist without being linked to a valid TeamID).
- **CRUD Functionality:** Users will be able to Create, Read, Update, and Delete their own scouting notes.

Functional Success:

- **Real-time Data Sync:** The application should successfully pull regional Teams and Events within a reasonable amount of time after they are published to the FTC API.
- **Display Teams and Notes:** Users must be able to view a team list by "Team Number" and a notes list by date order.
- **Mobile Responsiveness:** The UI must be fully functional on a mobile or tablet screen (common in competition pits), with no overlapping text or responsive issues.
- ***Stretch Goals:**
 - **Live Score Updates:** Pull data regularly about events, to make sure data is updated and current
 - **Media Management:** Successfully upload a robot photo from the client, store it in a media server (Cloudinary or imgBB), and display that image on the team's profile page.

Performance and Reliability:

- **Error Handling:** Be able to handle if the FIRST Tech Challenge Event API is down (main data is cached), invalid team requests, etc. with informative error messages
- **Deployment:** The full-stack app will be deployed using GitHub and Render and be accessible via a public URL

API and Data:

- Use the **FIRST Tech Challenge Events API** to fetch a list of all the events in a region, as well as the teams attending specific regional or championship events. The FTC Events API (<https://ftc-events.firstinspires.org/services/API>) will be the foundational team and event data (More information here:
 - <https://ftc-events.firstinspires.org/api-docs/index.html>)
- **Populate Team and Event Data** for user to select from when creating scouting notes
- **User Feature:** Allow team members create a Note Scout account to add custom notes connected to the team profiles pulled from the API. It creates a "living" scouting sheet that updates as the API runs new events in a region.
 - Possibly have live updates with Scores and Ranking displayed
 - Possibly in the future photos could also be uploaded using a photo service like imgBB or cloudinary since the PSQL DB will only hold the image URL)

Schemas (Draft)

User Table

- id: UUID (Primary Key)
- username: String (Unique)
- password_hash: String (For Bcrypt)
- role: String ('scout', 'admin')

Teams Table

- team_number: Integer (Primary Key)
- display_name: String
- Rookie_year: number

Event Table

- event_code: String: (Primary)
- event_name: String
- event_date: Date

Scouting Notes Table

- id: UUID (Primary Key)
- team_number: Integer (Foreign Key -> Teams)
- scout_id: UUID (Foreign Key -> Users)
- event_key: String (Foreign Key -> From FTC API, e.g., "USNYLI")
- notes: Text
- created_at: Timestamp

SQL Draft:

```
CREATE TABLE event_codes (  
    event_code VARCHAR(50) PRIMARY KEY,  
    event_name VARCHAR(255) NOT NULL,  
    event_date DATE NOT NULL  
);  
  
CREATE TABLE teams (  
    team_number INTEGER PRIMARY KEY,  
    team_name VARCHAR(255) NOT NULL,  
    rookie_year INTEGER,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE TABLE users (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    username VARCHAR(50) NOT NULL UNIQUE,  
    password_hash VARCHAR(255) NOT NULL,  
    role VARCHAR(20) DEFAULT 'scout',  
    is_admin BOOLEAN NOT NULL DEFAULT FALSE,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE TABLE notes (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    team_number INTEGER NOT NULL REFERENCES teams(team_number) ON DELETE CASCADE,  
    scout_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
    event_key VARCHAR(50) NOT NULL,  
    note_text TEXT NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

User Flow:

Authentication Steps

- Landing Page: Welcome and App overview
- Login/Signup: Form for entering credentials
- Verification: For login, compare password hash in the PostgreSQL DB and returns a JWT
- Redirect: to Scouting Section

Scouting Steps

- Select an Event from a list populated from the FTC Events API
 - This will return a list of teams at that event, which then can be selected to review past notes and write new notes
- Select a Team from a list populated from the FTC Events API
 - This will bring you directly to the team page where you can review past notes and write new ones
- Display Info
- Fill out the Scouting Form with Text
 - Create - the current user (or admin) can create the note
 - Update (PATCH) - the owner of the note (or admin) can update the note
 - Delete - the owner of the note (or admin) can delete the note